Project 3: Real-time 2-D Object Recognition

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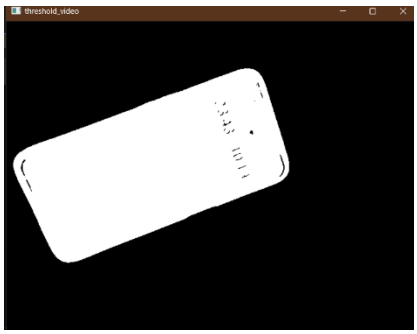
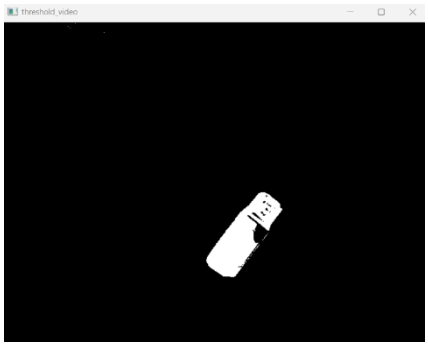
The project is about Real-time 2-D Object Recognition, It is a real-time recognition system, which classifies the objects based on their shape features. There are two modes for this project the first mode is training mode in which it allows the user to label the identified objects, and in normal mode it detects the objects using the training data. It uses two types of classifications, the first one is by using Euclidean distance metric and the other one is K-Nearest Neighbor matching.

Task-1 Threshold the input video

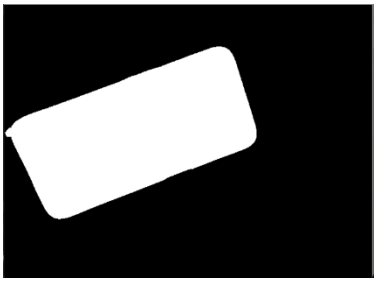
In Threshold the input video task, the program converts the video into grayscale and then converts into binary image by thresholding the video, The program uses the threshold function which picks the appropriate threshold value based on the input video and converts the pixels above the threshold as black and below the threshold as white.





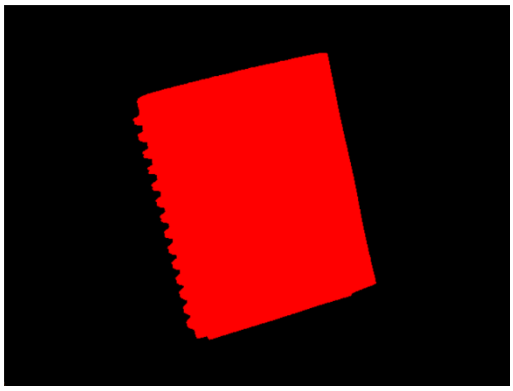
Task 2: Clean up the binary image

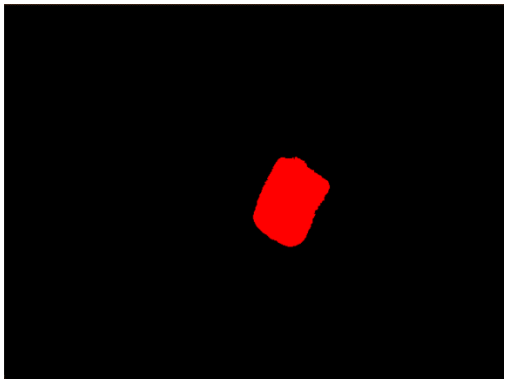
In the Clean up the binary image task, the program cleans the binary image and remove the defects from the image by using dilation and erosion methods



Task 3: Segment the image into regions

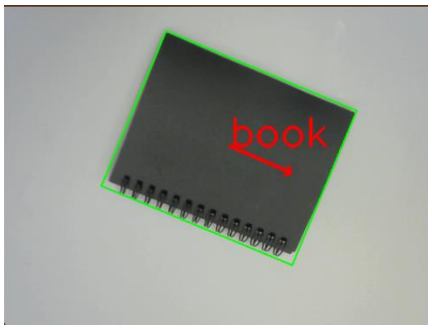
In the Segment the image into regions task, the cleaned binary image is segmented into different regions, this can be achieved by using inbuilt CV function connectedComponentsWithStats(). After finding all regions the program filters top 3 regions with area more than 3000 pixels. Then the program assigns different colors to the regions.





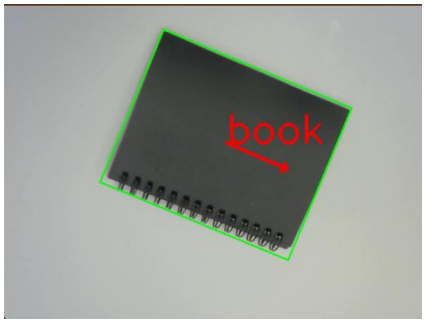
Task 4: Compute features for each major region

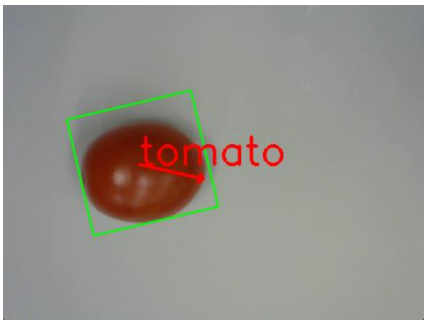
In the Compute features for each major region task, the program calculates different features of the identified region. Features such as axis of least moment of inertia, moments and HuMoments. These moments are invariant to scaling, translation and rotation.

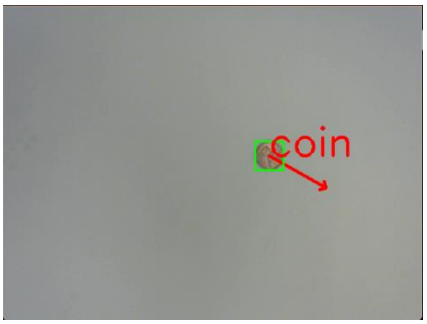


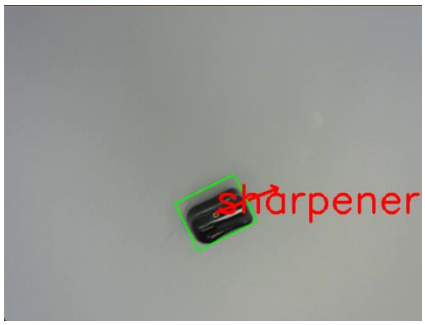
Task 5: Collect training data

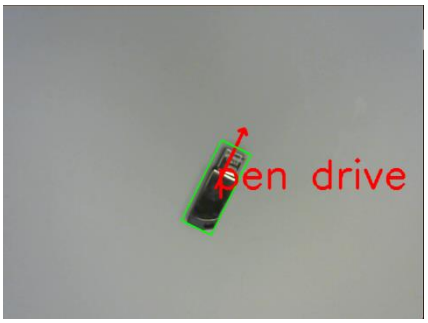
In the Collect training data task, the user has to set the program into training mode and has to give two arguments as inputs in which the first argument is the CSV file path and the other is ‘k’ or ‘n’ specifying the classifier. Then the user has to press different keys specifying the different names of the objects, the programs labels according and saves the data into CSV file,

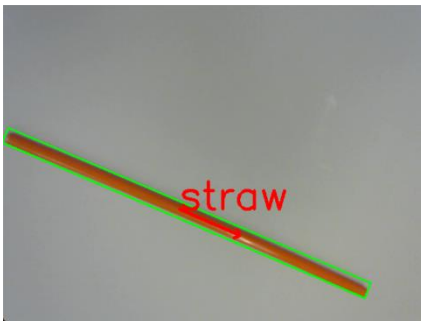












Task 6: Classify new images

Now set the program in inference mode in which the program identifies different objects by using the training database. It classifies objects by using Euclidean distance metric i.e. [(x_1 - x_2) / stdev_x].

Task 7: Implement a different classifier

Now set the program in inference mode in which the program identifies different objects by using the training database. It classifies objects by using K-nearest neighbor matching method(KNN).

Task 8: Evaluate the performance of your system

Here is the confusion matrix for measuring systems performance.

Reflections and Acknowledgement:

From this project, I learned How to access the images from a directory and store the feature vectors into a .csv file. I also learned how to detect objects by thresholding and finding moments. . As I wrote most of the code without using inbuilt functions it is very useful to get deeper knowledge on the Real-time 2-D Object Recognition.

When working on the assignment, I discussed with the TA on how to plot texture histograms and I got cleared.